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Characterization of Salt Precipitation Behavior in a Carbonate Formation Under Reservoir Conditions Using Low-Field Nuclear Magnetic Resonance

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Abstract

CO₂ geological storage in saline aquifers is considered as the most promising approach to reduce carbon emissions. The injected CO₂ will evaporate the brine, resulting in salt precipitation in the formation. Traditional experimental studies of salt precipitation at core-scale exhibit insufficient measurement accuracy and fail to quantitatively assess the impact of salt precipitation on the pore structure characteristics of core sample. Herein, we presented a work on the kinetics of salt precipitation during CO₂ geological storage in a typical low-permeability carbonate reservoir. Special attentions were given to the influences of pressure, flow rate, temperature and salinity, *etc.*, on salt precipitation at the core scale using Low-field NMR technology. The findings indicated that the temperature and salinity could greatly affect the salt precipitation since the maximum reduction in permeability reached 52.95%, while porosity reduction reached 28.48% at 200000 mg/L and 100 °C. The increase in flow rate (10 to 15 ml/min) might strengthen the ability of CO₂ to carry brine and mitigate the capillary back flow effect, thereby reducing the salt precipitation. The significant deviation in the one-dimensional signal spectrum indicated that salt precipitation predominantly occurred at the forefront of the core. Additionally, the injection pressure showed minimal influence on salt precipitation under varying salinity levels. Salt precipitation caused greater damage to the petrophysical properties of core with low-permeability. This study offered guidelines for mitigation strategies to minimize the adverse impacts of salt precipitation.

Keywords: CO₂ geological storage, Salt precipitation, Carbonate reservoir, in-situ NMR, Core scale injectivity

Introduction

Global energy demand has been growing rapidly due to the development of the world economy. Traditional energy sources such as coal, oil, and natural gas remain the primary energy sources worldwide, with their combustion producing large amounts of CO₂ and other greenhouse gases, leading to climate change. The data from the International Energy Agency indicated that global CO₂ emissions exceeded 368 billion tons

in 2022, marking a new record (Yang et al. 2023). CO₂ geological sequestration (CCS) serves as one of the most important measures in achieving global carbon neutrality by significantly reducing atmospheric CO₂ concentrations. Suitable storage sites for CO₂ include deep saline aquifers, abandoned oil and gas reservoirs, unrecoverable coal layers (Orivri et al. 2025, Eigbe et al. 2023). Deep saline aquifers are usually distributed throughout the sediment basins in mainland and overseas. These locations hold great potential in terms of storage capacity, duration, economic and environmental technical aspects (Wang et al. 2025, Kalyoncu Pakkaner et al. 2025).

The injection of dry CO₂ will evaporate the brine, resulting in the salt precipitation in the formation during the CCS operation. The salt precipitation during the CO₂ sequestration process has a significant impact on injection productivity, storage capacity, and long-term safety (Sun et al. 2025, Nasiri et al. 2025). This is due to salt crystals formed during precipitation blocking pore channels, hindering CO₂ flow and increasing injection pressure. Growing attentions have been given to the investigation of salt precipitation in recent years. For instance, the Berea sandstones ($K = 79\text{--}500$ mD) were used to conduct the coreflooding tests ($T = 35\text{--}50$ °C, $P = 6.3\text{--}8.2$ MPa) to study salt precipitation effects by Muller et al., Wang et al., and Ott et al. (Muller et al. 2009, Wang 2010, 2009, Ott 2010). Their results suggested that the absolute penetration rate reduced by 60 % while the CO₂ relative permeability reduced by 50 %, and the salt precipitation was observed at the inlet of the core samples. To further explore the salt precipitation behavior under high temperature condition, Ott et al., (Ott et al. 2013, Ott et al. 2021) and Tang (Tang et al. 2015) conducted the experimental studies under 110 °C. It was found that the precipitation state was affected by the injection rate. A low injection rate led to localized salt precipitation, while a higher injection rate resulted in uniform precipitation. The permeability and porosity of core samples decreased by 83.3 % and 14.6% under high temperature conditions. They also claimed that capillary pressure differences and brine salinity were identified as key controlling parameters to affect the salt precipitation. Akindipe et al. (Akindipe et al. 2022) studied the salt precipitation in sandstone and carbonate rock under 50 °C and 10.5 MPa. The results shown that the level of heterogeneity affected the precipitation initiation, its duration, and the amount of precipitates. Lower salt saturations were encountered in less heterogeneous sandstone compared to more heterogeneous carbonate. Heterogeneity also influenced the distribution of salt clusters within pore elements such that, for the same wettability state, medium-sized clusters dominated in sandstone, while large clusters were the main contributors to the total salt volume in more heterogeneous carbonate. Despite the fact that researches on salt precipitation during CO₂ injection have yielded many important results, however, there are fewer reports on the salt precipitation behavior in carbonate reservoirs under high temperature, high pressure, and high salinity conditions. Furthermore, traditional experimental studied of salt precipitation at core scale showed insufficient measurement accuracy and could not quantitatively assess the impact of salt precipitation on the pore structure characteristics. Therefore, we carried out a study on the kinetics of salt precipitation during CO₂ geological storage in a typical low-permeability carbonate reservoir. Special attentions were given to the influences of pressure, flow rate, temperature and salinity, *etc.* on salt precipitation at core scale using Low-field NMR technology. This study enhances the fundamental understanding of the mechanisms governing salt precipitation during CO₂ injection in HT-HP carbonate reservoirs, offering operational guidance for field-scale implementation.

Experimental section

Materials

The brine used in this work was formulated based on the composition tests of formation water, as presented in Table 1. To further investigate the influence of salinity, the initial salinity was diluted to 179508 mg/L and 119672 mg/L, respectively. The core samples used in this work were collected from a carbonate reservoir in the Middle East, and the petrophysical properties of the core plugs are shown as Table 2. CO₂ with a purity

of 99.99% was supplied by TianJin Gases Co. Ethanol (99.8%, AR) and deionized water were purchased from Beijing Tongguang chemicals Co.

Tabel 1—Composition and proportion of brine (Total salinity: 200000 mg/L)

Agent	SrSO ₄	NaCl	Na ₂ SO ₄	NaHCO ₃	CaCl ₂	MgCl ₂	KCl
Content (g/L)	0.548	120.18	52.46	12.9	17.67	8.42	2.27

Tabel 2—Petrophysical properties of the core plugs

Core sample	Diameter (cm)	Length (cm)	<i>K</i> (mD)	ϕ (%)
#1	2.49	5.01	60.02	20.82
#2	2.50	5.00	37.48	19.97

Core samples preparation

The core samples were firstly dried in an oven at 100 °C for 7 d to eliminate any moisture. The dried core was then subjected to a vacuum (ensuing no air was trapped in the core) for 2 h and then filled with ethanol for 3 h. Afterwards, the ethanol saturated core was placed horizontally into a coreholder to conduct ethanol injection from 0.5-6.0 ml/min at room temperature, through which the core permeability (*K*), pore volume (by mass balance), and porosity can be determined.

Online NMR coreflooding measurements

The initial pore characteristics (CPMG and HSE sequence) were measured with ethanol saturated state by NMR system (Figure 1). The core was dried at 100 °C for 5 h and then saturated with brine by vacuuming procedure for 3 h. The brine saturated core was then placed into the NMR system, and slowly pressurized to the specified pressure and temperature. Once the pressure and temperature reached equilibrium, CO₂ was injected into the system at different constant rate. CO₂ injection would stop when the fluid signal in the system stabilized, and the core swept by CO₂ would be dried at 100 °C for 24 h to remove residual water. Subsequently, the dried core with salt precipitation was saturated with ethanol to measure the pore characteristics, permeability and porosity. The core was eventually cleaned by deionized water at 100 °C for 3 d and applied to other cases (Table 3).

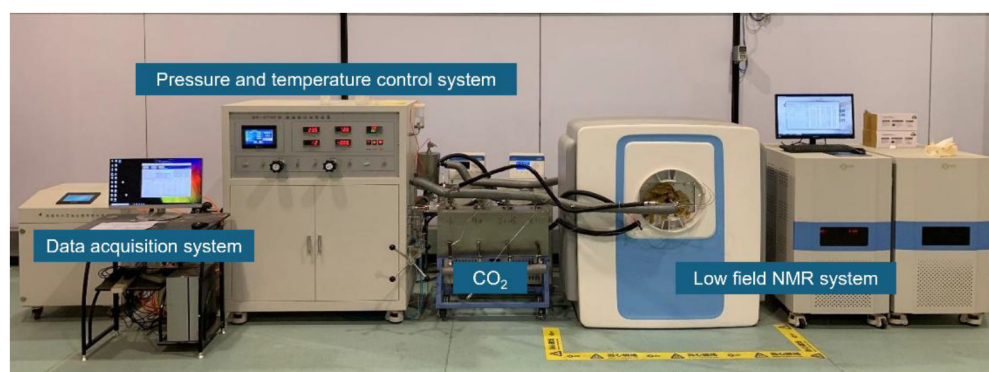


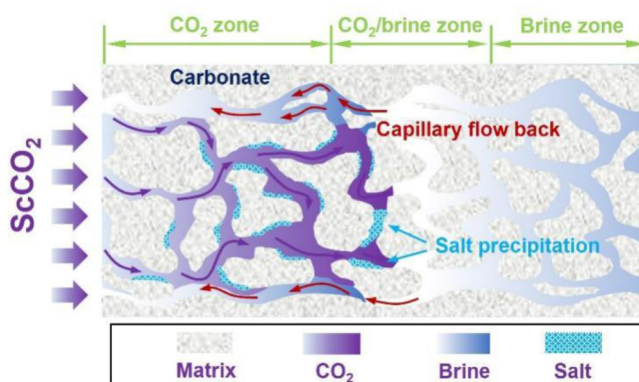
Figure 1—HT-HP online NMR coreflooding system

Table 3—Scenarios of the tests in the CO₂ injection experiment

Tests	Salinity (mg/L)	K (mD)	P (MPa)	v (ml/min)	T (°C)
1	119672	60.02	14	10	50
2	119672	60.02	14	15	50
3	119672	60.02	14	10	100
4	179508	60.02	14	10	50
5	200000	60.02	14	10	50
6	200000	60.02	14	10	100
7	200000	60.02	24	10	100
8	119672	60.02	14	10	80
9	119672	37.48	14	10	80

Results and discussion

The interaction between CO₂ and brine can be divided into three typically regions: CO₂ zone, CO₂-brine zone and brine zone (Figure 2). The CO₂ zone refers to the areas of intense interaction between CO₂ and brine. When the CO₂ was injected into the formation, it dissolved in the brine and induced significant evaporation of water from brines due to the presence of dry CO₂. These effects resulted in an increase in HCO₃⁻ and salinity of brine, ultimately leading to salt precipitation (Ghaedi et al. 2025). The CO₂-brine zone and brine zone were defined as the minor and non-interacting areas. The brine in the CO₂-brine zone would flow back to CO₂ zone due to capillary force when it was fully evaporated by CO₂. This phenomenon, known as capillary flow back effect would strengthen the salt precipitation in the CO₂ zones. By analyzing the pore characteristics (T₂ map and one-dimensional signal spectrum) of rock cores before and after gas injection, this study investigated how various factors affected salt precipitation and further explored their impact on the physical properties of rock core.

Figure 2—The diagram of the interaction between CO₂ and brine in the formation

Influence of injection rate

Increasing the flow rate of CO₂ can enhance its interaction with brine. While our results shown that there were no significant changes in the total signals of T₂ map and one-dimensional signal spectrum when the injection rate increased from 10 ml/min to 15 ml/min (Figure 3a and 3b). This indicated that the increasement of flow rate did not lead to an increase in salt precipitation. The permeability and porosity even increased when the injection rate was at 15 ml/min (Figure 3c and 3d). Actually, increasing the flow rate of CO₂ can inhibit the capillary flow back effect, which lowed the salt precipitation in the CO₂ zone. Moreover, in actual construction, increasing flow speed also entails corresponding costs and safety hazards.

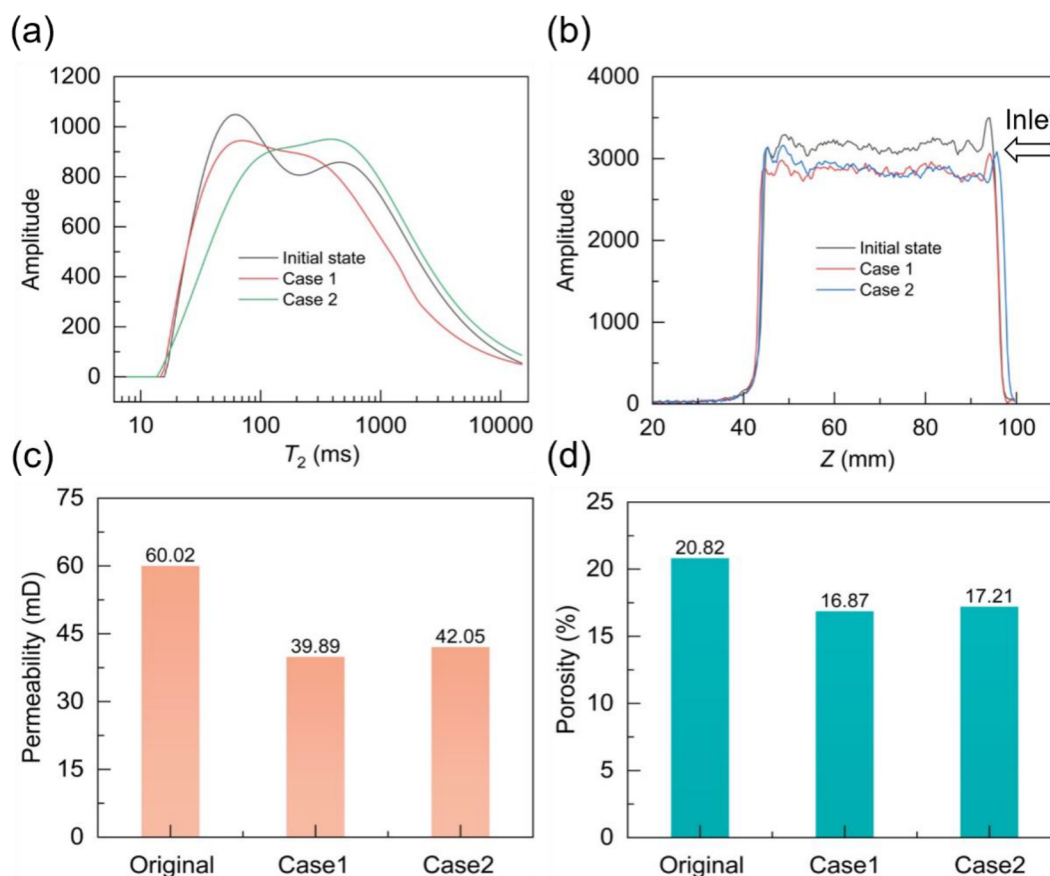


Figure 3—The influence of injection rate of CO₂ on the salt precipitation. (a) T₂ map, (b) one-dimensional signal spectrum, (c) permeability and (d) porosity of the core sample after the injection of CO₂ at different flow rate.

Influence of temperature

During the process of CO₂ injection into formation, temperature changes significantly impact salt precipitation behavior. Specifically in high-salinity reservoirs, increased temperature leads to enhanced salt precipitation due to the thermal expansion effect on brine saturation (Cheng et al. 2025). Our research findings indicate that when temperature increases from 50 °C to 100 °C, both the T₂ spectrum and one-dimensional signal spectrum decreased significantly (Figure 4a and 4b). Specifically, the T₂ spectrum gradually transitioned to a single-peak state, reflecting an increase in salt precipitation. The permeability and porosity decreased by 44.27% and 24.88% after CO₂ injection at 100 °C (Figure 4c and 4d). Although increasing temperature reduced CO₂ dissolution in brine, higher temperature enhanced the possibility of brine remaining in situ and evaporating locally (Zhang et al. 2025).

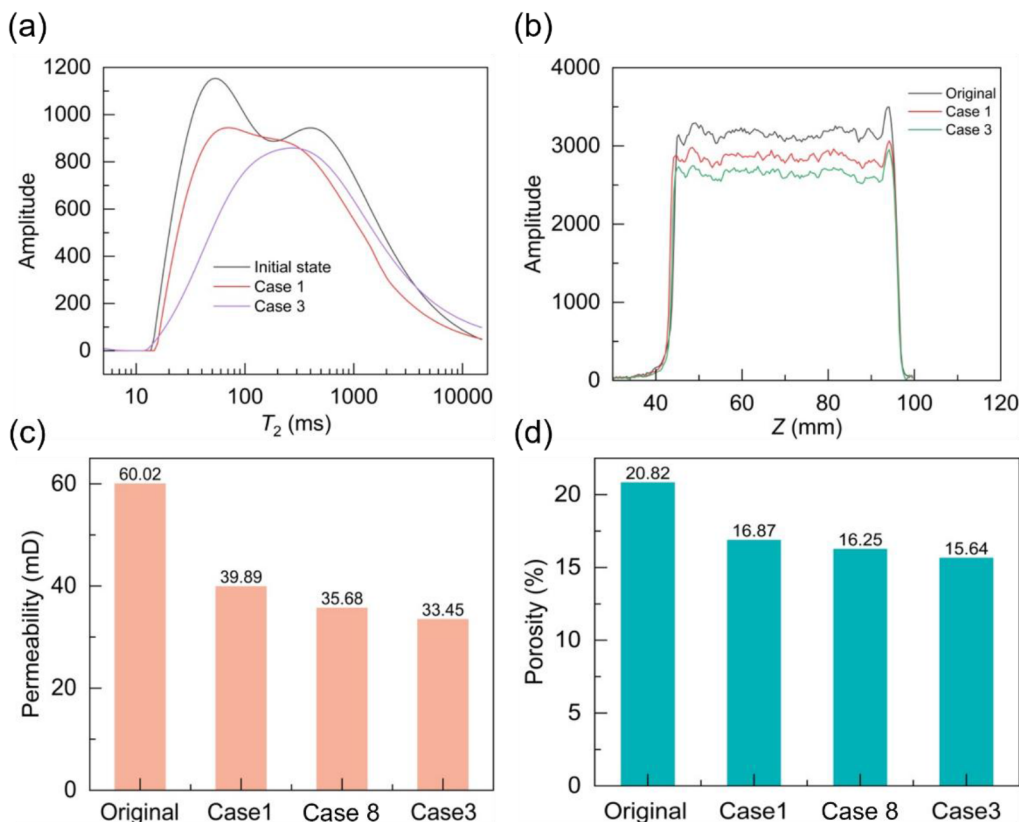


Figure 4—The impact of temperature on salt precipitation. (a) T₂ map, (b) one-dimensional signal spectrum, (c) permeability and (d) porosity of the core sample after the injection of CO₂ at different temperature.

Influence of salinity

The amplitude of T₂ map and one-dimensional signal spectrum was continuously decreased with increasing salinity (Figure 5a and 5b). The signal offset in the inlet of the core was more notable (Figure 5b), which demonstrated that the salt precipitation at the injection side of the core was more severe. This phenomenon was closely related to the capillary flow back effect that the brine in the back side mitigated toward to the injection side. Furthermore, the permeability and porosity of the core decreased 52.95% and 28.48% when the salinity increased to 200000 mg/L. Surface microscopic images of the core revealed that the crystallite morphologies predominantly exhibited diamond-shaped structures with dispersed distribution under moderate mineralization levels (119672 mg/L) and 50 °C. As the temperature rose to 100 °C, the crystallites aggregated toward the flow channels and pores. The crystallites presented radiating cluster-like distributions in a 200000 mg/L level and at 100 °C. Through X-ray diffraction (XRD) analysis, it was found that the primary components of the salt crystals were mainly sodium chloride with minor amounts of calcium sulfate. This indicated that under these specific conditions, the crystals formed a distinct pattern rather than clustering near particular structures like flow channels and pores, as observed at lower temperature.

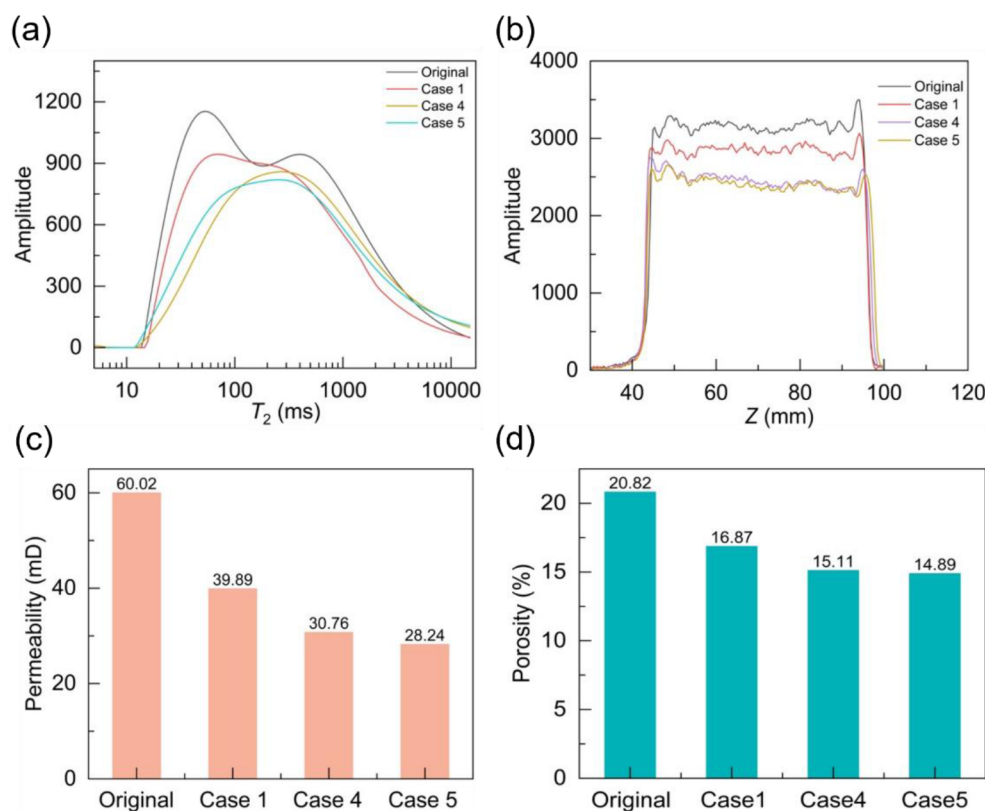


Figure 5—The impact of salinity on salt precipitation. (a) T_2 map, (b) one-dimensional signal spectrum, (c) permeability and (d) porosity of the core sample after the injection of CO_2 at different salinity.

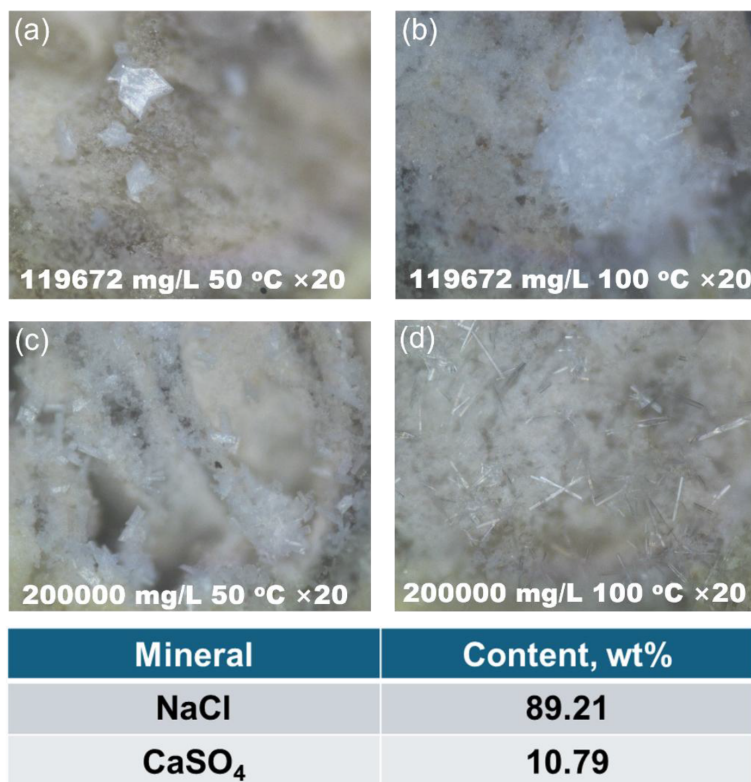


Figure 6—The influence of temperature and salinity on the morphology of salt crystals. micrograph of salt crystals at (a) 119672 mg/L and 50 °C, (b) 119672 mg/L and 100 °C, (c) 200000 mg/L and 50 °C, (d) 200000 mg/L and 100 °C.

Influence of injection pressure

Pressure may have an impact on the solubility of CO₂ in brine and influence its precipitation behavior. It was found that the T₂ map and one-dimension signal spectrum basically had no change when injection pressure increased 10 MPa (Figure 7a and 7b). Same results were observed in the measurements of permeability and porosity (Figure 7c and 7d). These were mainly attributed to the following reasons. The solubility of CO₂ in brine had reached equilibrium at the pressure of 14 MPa (Zhang et al. 2018). Additionally, the differential pressure in Case 6 demonstrated no significant difference compared to Case 7, which resulted in minor changes of salt precipitation. Therefore, it is essential to apply the necessary pressure to minimize the increased costs associated with raising pressure in CCS operations.

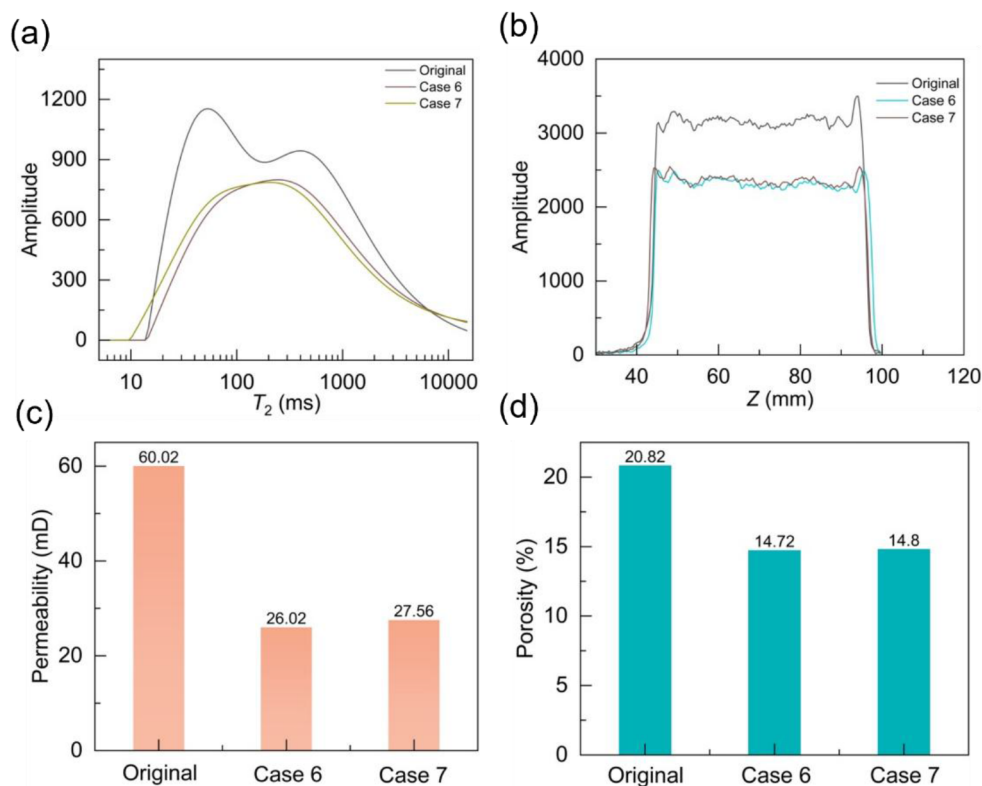


Figure 7—The impact of injection pressure on salt precipitation. (a) T₂ map, (b) one-dimensional signal spectrum, (c) permeability and (d) porosity of the core sample after the injection of CO₂ at different pressure.

Influence of pore structure

To investigate the impact of pore structures on salt precipitation, a core sample with lower permeability (37.48 mD) and an initial T₂ spectrum showing a single-peak distribution was prepared. The results indicated that the one-dimensional signal spectrum of the core revealed significant fluctuations compared to the initial state and became more uniform after CO₂ injection (Figure 8b). The T₂ spectrum shifted to the right (Figure 8a), demonstrating that the salt precipitation led to a more uniform pore structure of the core. Additionally, the variations in permeability and porosity of the rock core exceeded those observed in high-permeability core sample (60.02 mD) under the same conditions (Figure 8c), suggesting that salt precipitation caused greater damage to the petrophysical properties of core with low-permeability.

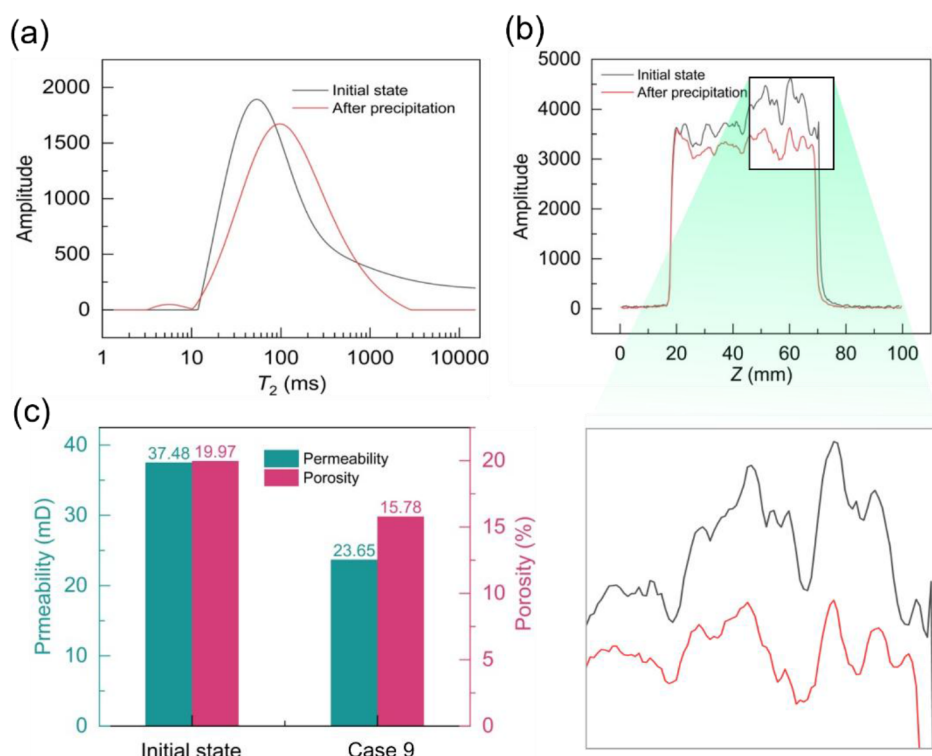


Figure 8—The impact of pore structure on salt precipitation. (a) T₂ map, (b) one-dimensional signal spectrum, (c) permeability and porosity of the core sample after the injection of CO₂.

Conclusion

In order to investigate the salt precipitation behavior during the CO₂ geological storage process in high-mineralization carbonate reservoirs under high pressure and temperature conditions, this paper employed low-field NMR technology to explore the influence of injection rate, pressure, and temperature, *etc.* on salt precipitation. Based on the data, the following conclusions can be generally drawn:

1. The injection rate and pressure had little impact on the salt precipitation, with increase in injection speed and pressure resulting in only a 5.9% and 2% rise respectively in rock core permeability and porosity.
2. The temperature and salinity could greatly affect the salt precipitation since the maximum reduction in permeability reached 52.95%, while porosity reduction reached 28.48% at 200000 mg/L and 100 °C.
3. Salt precipitation caused greater damage to the petrophysical properties of core with low-permeability.
4. For the field application of CCS, it is important to balance the necessary pressure and flow rate in order to minimize increased costs associated with raising pressure while effectively controlling salt precipitation during the CO₂ injection.

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